collators.py — technical

`src/moe_pruning/data/collators.py · 92 lines · moe_pruning.data.collators`

Purpose

Padding and per-stack masks. Everything is right-padded.

Step by step

Classes and functions in source order. Signatures, line ranges, raised exceptions and settings are extracted from the code; numbered steps describe what the code does.

`class ModelBatch`

lines 24-41 · dataclass

Padded batch with per-stack real-token masks.

Field	Type	Default
`example_ids`	list[int]	
`input_ids`	torch.Tensor	
`attention_mask`	torch.Tensor	
`labels`	torch.Tensor	
`masks`	dict[str, torch.Tensor]	field(default_factory=dict)

`ModelBatch.model_inputs`

```
def model_inputs(self, device: torch.device) -> dict[str, torch.Tensor]
```

lines 31-36

input_ids, attention_mask and labels moved to a device.

1. Return `input_ids`, `attention_mask`, `labels` moved to the given device.

`ModelBatch.token_masks`

```
def token_masks(self, device: torch.device | None=None) -> dict[str, torch.Tensor]
```

lines 38-41

Real-token mask of each stack (encoder / decoder).

`_pad`

```
def _pad(items: Sequence, attr: str, value: int) -> torch.Tensor
```

lines 48-54

1. Allocate `[B, max_len]` long tensor filled with the pad value; copy each sequence into its row prefix.

`collate_seq2seq`

```
def collate_seq2seq(items: Sequence, pad_token_id: int) -> ModelBatch
```

lines 57-69

Encoder-decoder batch.

1. `input_ids` padded with the tokenizer pad id; `attention_mask[i, :len] = 1`.
2. `labels` padded with -100 (ignored by cross-entropy).
3. Masks: `encoder = attention_mask.bool()` ([B, S_enc]); `decoder = labels != -100` ([B, S_dec]). Because the decoder input is the right-shifted labels, position j is real exactly when label j is real.

`collate_causal`

```
def collate_causal(items: Sequence, pad_token_id: int) -> ModelBatch
```

lines 72-86 · raises ValueError

Decoder-only batch: `labels` has the same length as `input_ids`.

1. Require `len(labels) == len(input_ids)` for every item (labels aligned with inputs; the model shifts).

2. Pad inputs and labels as above; single mask `decoder = attention_mask.bool()` - prompt and response tokens are real for MoE statistics even if their labels are -100.

`iter_batches`

```
def iter_batches(items: Sequence, batch_size: int, pad_token_id: int) -> Iterator[ModelBatch]
```

lines 89-92

Encoder-decoder batches (kept for compatibility; prefer `task.iter_batches`).

Module constants

Name	Value
`LABEL_PAD`	-100

Errors raised

Function	Exception	Message (f-string fields in braces)
`collate_causal`	ValueError	Causal examples need labels aligned with input_ids